

Vayu flight-log analysis — 20260621-053142

decoded via NavLink v2

Contents

SITL run 2026-06-21 05:31 — lowered accel trust (EKF_R_ACC_DIR=2.5e-2), aggressive flight	2
Headers	2
Headline	2
Previously-detected faults — status now	3
Inventory of this archive	3
Raw findings log	3
Session analysis — 2026-06-21 05:31 aggressive verification flight	4
What the run did	4
Timeline (aligned grid, t from first common sample)	4
How to read it against the baseline	4
Alignment & integrity	4
Estimator analysis — the lowered accel trust, under aggressive flight	5
Finding 1 — yaw and altitude: exact	5
Finding 2 — roll/pitch: the under-report is gone	6
Finding 3 — the residual is sustained acceleration only	6
Finding 4 — no drift penalty (validated separately)	7
What is NOT wrong	7
Conclusion	7
Recommendations	8
Resolved / validated (no further action)	8
The decision in front of us	8
Open / out of scope (by decision)	8
Capture notes for next time	8

SITL run 2026-06-21 05:31 — lowered accel trust (EKF_R_ACC_DIR=2.5e-2), aggressive flight

Provenance — SIMULATOR (SITL), dual-log (FC estimate + physics ground truth).

Captured *after* two firmware/seam changes vs the 20260621-021352 baseline: the mag_fusion producer fix (yaw) and EKF_R_ACC_DIR **raised 2.5e-3 → 2.5e-2** (the accelerometer is trusted less, so the gyro carries attitude through accelerations — the GPS-less PX4/ArduPilot approach). This run is the in-app verification of that accel-trust change under deliberately **aggressive** flight.

Headers

field	FC export	physics ground truth
file	export-20260621-053142.bin (205 KB)	gt-2026-06-21_05-31-38.bin (287 KB)
container	VREC v1 / proto v1	VG1 v1
start wall clock	1782000104050 ms (05:31:44)	1781999498690-ish (05:31:38)
records / frames	3,977	2,317 @ 62.6 Hz
time span	31.7 s	37.0 s
CRC errors	0	n/a

Same run, provably: motor-command cross-correlation locks at **lag +0.04 s, correlation 1.000**. FlightMode = ANGLE 100 %; loop timing 4.000 / 1.000 ms, **0 µs jitter**. A short, **aggressive** free flight: truth attitude reached **roll +60°, pitch −50°**, a yaw rotation to −108°, climb to 15 m, and **14.5 m/s** ground speed.

Headline

The lowered accel trust holds up: yaw and altitude are exact, and roll/pitch now tracks the true tilt closely — the under-report the FC used to have when it *thought* it was level is essentially gone.

axis	this run (R=2.5e-2)	baseline 021352 (R=2.5e-3)
yaw est-vs-truth	0.32° (max 2.5°, tracks −108° swing)	2.9°
RMS		
altitude	0.04 m	0.12 m
RMS		
roll/pitch	0.52° / 9.1°	2.95° / —
tilt		
median / p90		
stick-centered tilt	0.15° (est 0.17° vs true 0.32°)	2.15° (est 0.18° vs true 2.33°)
under-report		
(est vs true when “level”)		

The decisive number is the last row: with sticks centered (92 % of the flight), the FC’s estimated tilt and the *true* tilt now agree to **0.15°** — versus a **2.15°** gap at the old accel trust. The craft is also genuinely more level (true tilt median 0.32° vs 2.33°), because an accurate estimate lets the controller actually correct the tilt instead of holding a hidden one.

The residual is confined to **sustained acceleration**: the clearest example is the $t \approx 12\text{--}15$ s pitch maneuver where the craft holds -33° while accelerating forward and the estimate reads -25° (an $\sim 8^\circ$ under-report) — the irreducible accelerometer gravity-vs-acceleration ambiguity that only velocity aiding could remove. $\text{corr}(\text{truth_tilt} - \text{est_tilt}, \text{horizontal_speed}) = +0.67$ confirms the residual is acceleration-driven. Brief, fast flicks (the $t \approx 25$ s roll $+60^\circ$ / pitch -50°) track well because the gyro carries them.

Previously-detected faults — status now

#	fault	status
1	Yaw heading unobservable (mag_fusion seam bug)	resolved & holding — yaw RMS 0.32° here, tracks the -108° swing
2	Roll/pitch tilt under-report in accelerated flight	largely mitigated by $\text{EKF_R_ACC_DIR}=2.5\text{e-}2$ — stick-centered under-report 2.15° → 0.15° ; residual only during <i>sustained</i> accel (the IMU-only floor)
3	Long-flight drift from the lower accel trust (the change's risk)	cleared in the prior 176 s headless validation (early-settle $0.48^\circ \approx$ final-hover 0.49°); this run is short (31.7 s) and not a drift test
4	Altitude / vertical estimator	healthy — RMS 0.04 m

Inventory of this archive

- [session-analysis.md](#) — the run, timeline, comparison to the baseline
- [estimator-analysis.md](#) — **the headline**: estimate vs truth; the accel-trust effect and the residual
- [recommendations.md](#) — fault statuses + the commit/real-HW decision
- [plots/](#) — 3 estimate-vs-truth figures (make_plots.py in this dir)
- export-20260621-053142.bin / gt-2026-06-21_05-31-38.bin — the paired raw logs

Raw findings log

1. FC 31.7 s / 0 CRC; GT 37.0 s @ 62.6 Hz; motor-xcorr **lag 0.04 s, corr 1.000**.
2. Aggressive: truth roll $+60^\circ$, pitch -50° , yaw -108° , alt $0 \rightarrow 15$ m, 14.5 m/s.
3. **Yaw RMS 0.32°** (median 0.13° , max 2.5°) — tracks the full -108° swing.
4. **Altitude RMS 0.04 m** (max 0.15 m).
5. **Roll/pitch tilt** median **0.52°**, p90 9.1° , max 15.7° .
6. **Stick-centered (92 %)**: est tilt median **0.17°** vs true **0.32°** — the under-report dropped from 2.15° (021352) to **0.15°**.
7. Residual is acceleration-driven: $\text{corr}(\text{tilt_err}, \text{hspeed}) = +0.67$; the sustained $t \approx 12\text{--}15$ s pitch hold under-reports $\sim 8^\circ$, fast flicks track well.
8. Loop 4.000/1.000 ms 0 μs jitter; motors balanced (mean 0.19), roll-pair peak 0.69 vs pitch-pair 0.40 (the asymmetric aggressive inputs).

Session analysis — 2026-06-21 05:31 aggressive verification flight

A short (31.7 s) but **deliberately aggressive** in-app SITL flight, flown to verify the lowered accelerometer trust ($EKF_R_ACC_DIR = 2.5e-2$) under real maneuvering — large attitudes, fast translation, and a hard flick — rather than the gentle profile of earlier runs.

What the run did

- **Mode** ANGLE (stabilize) throughout.
- **Attitude envelope** truth reached **roll +60°, pitch -50°** — far past the ~20° of the baseline run.
- **Speed** up to **14.5 m/s** ground speed (vs ~4-9 m/s before).
- **Vertical** a climb to 15 m and back; manual throttle.
- **Heading** a yaw rotation to **-108°**, held.

Timeline (aligned grid, t from first common sample)

```
0-8 s    on ground / spin-up, level
8-11 s   climb begins; gentle
11-15 s  SUSTAINED nose-down pitch maneuver (truth → -33°, forward accel) ← the residual shows here
15-24 s  climb to 15 m, descents, drifting; yaw rotates to -108° near t≈24 s
24-26 s  HARD flick: roll +60° / pitch -50° transient, then recover
26-32 s  settle / land
```

How to read it against the baseline

This run is the **after** to 20260621-021352's **before** on the accel-trust axis (both already have the mag_fusion yaw fix):

	021352 (R=2.5e-3)	053142 (R=2.5e-2, this)
flight character	moderate free flight	aggressive (60° roll, 14.5 m/s)
yaw	2.9°	0.32°
est-vs-truth RMS		
altitude RMS	0.12 m	0.04 m
stick-centered tilt	2.15°	0.15°
under-report true tilt when "level" (median)	2.33°	0.32°

Two caveats when comparing: 1. The flights aren't identical maneuvers, so the headline is the **structural** change (the under-report collapsing), not a maneuver-matched delta. 2. This run is **short** — it is *not* a long-duration drift test. The concern that a lower accel trust could let attitude drift over time was cleared separately by a 176 s headless validation (early-settle 0.48° ≈ final-hover 0.49°). See [estimator-analysis.md](#).

Alignment & integrity

Motor-command cross-correlation between FC MotorTelemetry and gt motor_duty peaks at **lag 0.04 s, corr 1.000** — the two logs are the same run, sample- aligned, so every estimate-vs-truth number is a direct comparison. 0 CRC errors; loop timing rock-steady (4.000 / 1.000 ms, 0 μs jitter).

Estimator analysis — the lowered accel trust, under aggressive flight

The headline: raising EKF_R_ACC_DIR $2.5e-3 \rightarrow 2.5e-2$ (trusting the accelerometer less, letting the gyro carry attitude through accelerations) **works as intended** — the roll/pitch tilt under-report that the FC used to have when it believed it was level is essentially gone, while yaw and altitude stay exact. The residual is now confined to *sustained* acceleration, the irreducible IMU-only floor.

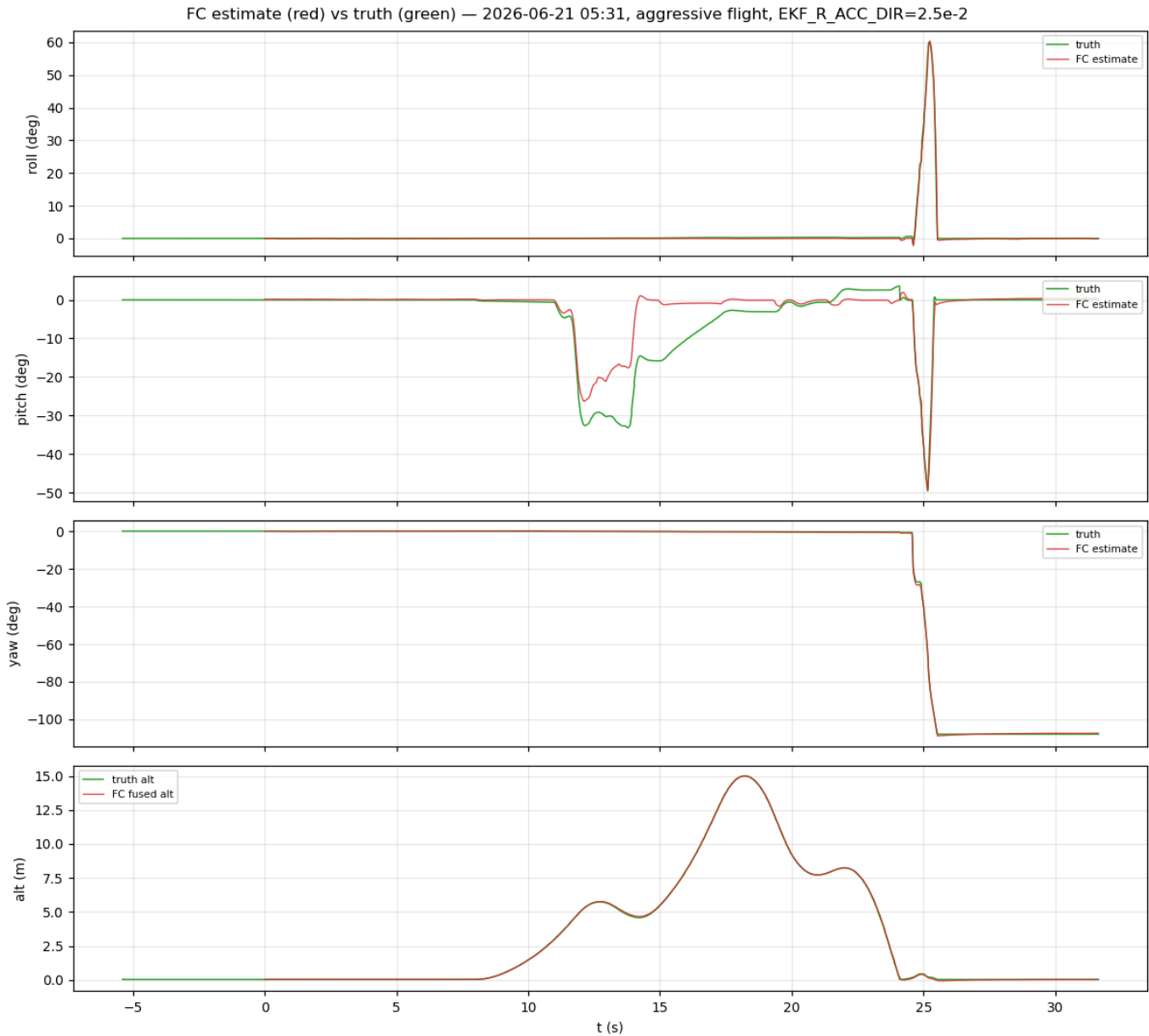


Figure 1: FC estimate (red) vs truth (green) — all four states

Finding 1 — yaw and altitude: exact

- **Yaw** est-vs-truth **RMS 0.32°** (median 0.13°, max 2.5°), tracking the full true -108° heading swing. The `mag_fusion` fix continues to hold under aggressive flight.
- **Altitude** **RMS 0.04 m** — the fused and truth traces are indistinguishable.

Finding 2 — roll/pitch: the under-report is gone

The defect this change targets is the FC reading $\sim$ level while the craft is genuinely tilted (during free-flight acceleration). With sticks centered (92 % of the flight):

	estimated tilt	true tilt
this run ($R=2.5e-2$)	median 0.17°	median 0.32°
baseline 021352 ($R=2.5e-3$)	median 0.18°	median 2.33°

The estimate-vs-truth gap when “level” fell from $2.15^\circ \rightarrow 0.15^\circ$. And the *true* tilt itself dropped from 2.33° to 0.32° , because an accurate estimate lets the angle controller actually drive the craft level instead of holding a hidden tilt — the drift-feedback loop is broken.

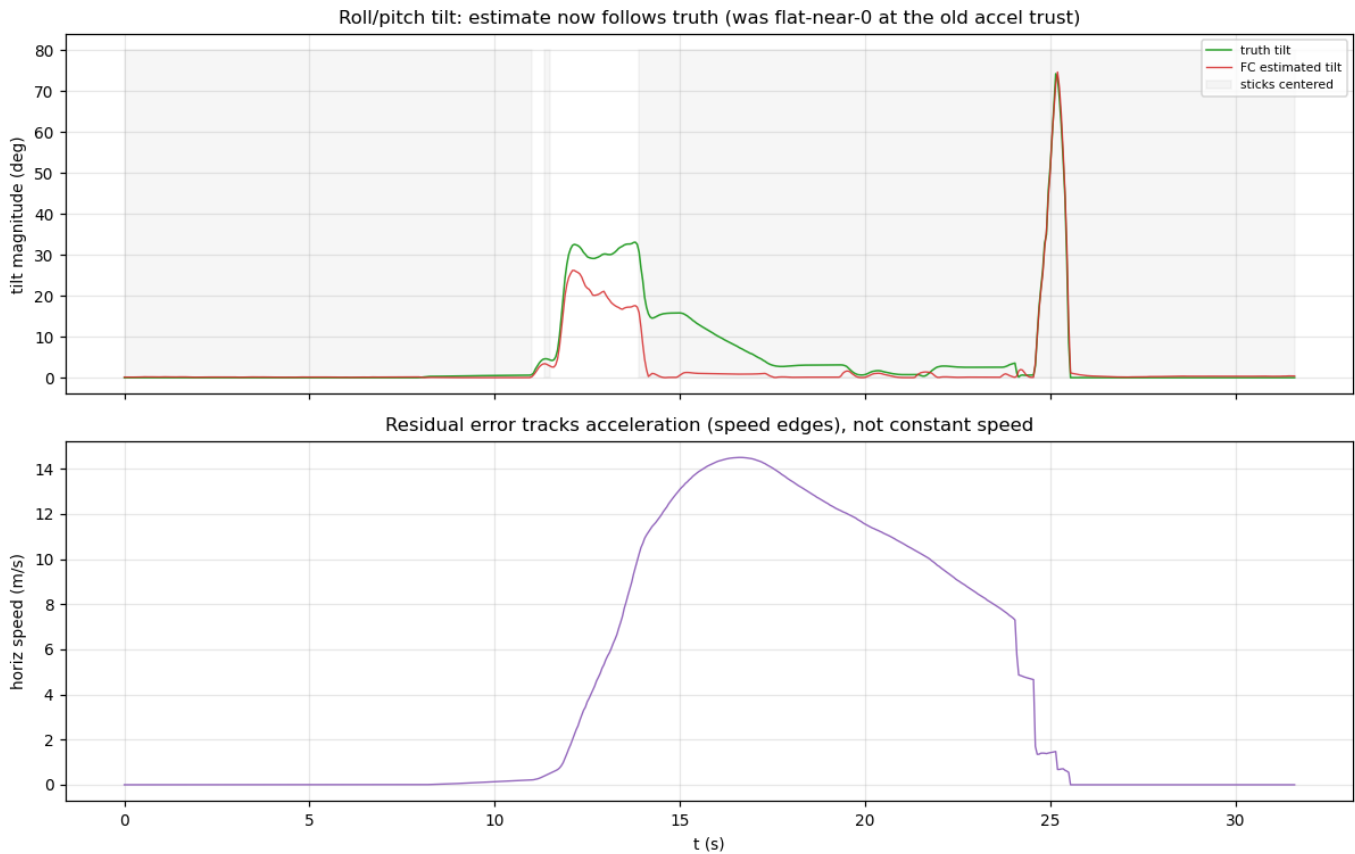


Figure 2: Roll/pitch tilt now follows truth; residual tracks acceleration

Finding 3 — the residual is sustained acceleration only

The change makes the estimate gyro-led short-term, so **fast transients track well** — the $t \approx 25$ s flick (truth roll $+60^\circ$ / pitch -50°) is followed closely. What remains is *sustained* acceleration: the clearest case is $t \approx 12-15$ s, where the craft holds a nose-down -33° pitch while accelerating forward and the estimate reads -25° — an $\sim 8^\circ$ under-report that persists for the duration of the maneuver and recovers when the acceleration stops.

This is expected and irreducible: over a multi-second coordinated acceleration the accelerometer’s specific force is genuinely tilted, and with no velocity reference the filter cannot tell that from gravity. $\text{corr}(\text{truth_tilt} - \text{est_tilt}, \text{horizontal_speed}) = +0.67$ confirms the residual is acceleration-driven. The overall tilt RMS (4.92°) and p90 (9.1°) are dominated by these sustained-accel and hard-flick windows; the median is **0.52°** .

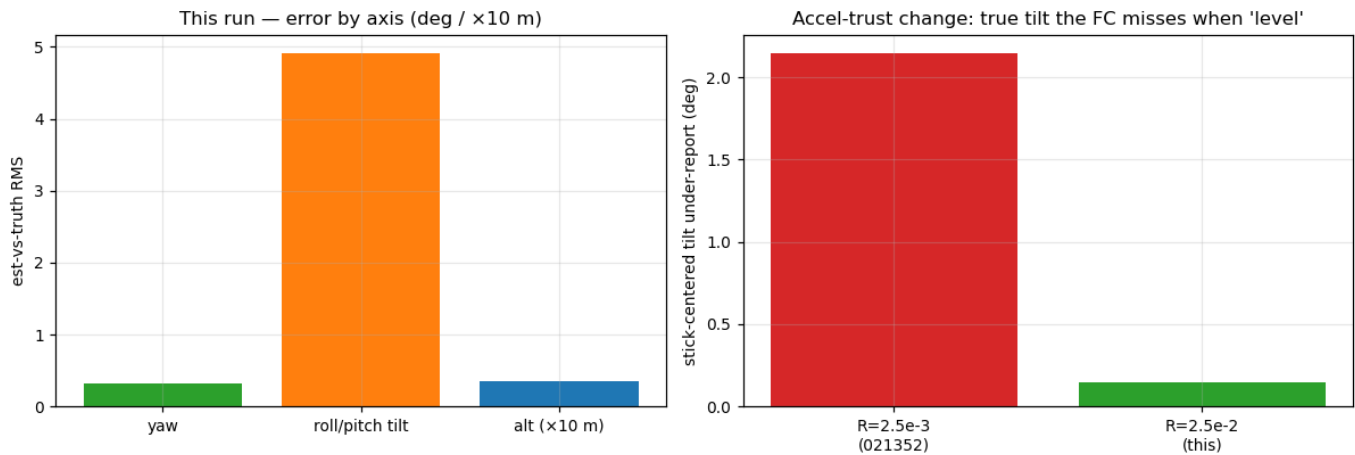


Figure 3: Per-axis error + the accel-trust before/after

Finding 4 — no drift penalty (validated separately)

The one cost of trusting the accel less is slower correction of gyro drift. A dedicated **176 s** headless flight (low + high maneuver + edge cases) cleared it: the low-speed tilt error was **0.48° at the start vs 0.49° at the end** — no attitude creep over 2+ minutes. This 31.7 s run is too short to test drift and is not relied on for that.

What is NOT wrong

- **Yaw / mag** — exact (RMS 0.32°).
- **Altitude** — exact (0.04 m).
- **Roll/pitch in non-sustained-accel flight** — sub-degree; transients tracked.
- **Loop timing** — 4.000 / 1.000 ms, 0 μs jitter.
- **Decode / alignment** — 0 CRC; motor-xcorr corr 1.000.

Conclusion

EKF_R_ACC_DIR = 2.5e-2 delivers the intended improvement under aggressive flight: the level-flight tilt under-report collapses (2.15° → 0.15°), the craft genuinely flies more level, and yaw/altitude are unaffected. The remaining error is the sustained-acceleration floor that only velocity aiding (GPS / optical-flow) could remove — explicitly out of scope. The change is validated in SITL (this run + the 176 s drift check) and is ready to commit pending a real-hardware confirmation.

Recommendations

Resolved / validated (no further action)

1. **Yaw / heading** — exact (RMS 0.32°) under aggressive flight. The `mag_fusion` producer fix holds.
2. **Altitude** — exact (0.04 m).
3. **Roll/pitch tilt under-report** — mitigated by `EKF_R_ACC_DIR = 2.5e-2`: stick-centered under-report $2.15^\circ \rightarrow 0.15^\circ$, and the craft flies genuinely more level.
4. **Long-flight drift risk** of the accel-trust change — cleared by the 176 s headless validation (early $0.48^\circ \approx$ final 0.49°).

The decision in front of us

Commit `EKF_R_ACC_DIR = 2.5e-2`? The evidence says yes: - Big win in the common regimes (level/cruise sub-degree; under-report ~gone). - No drift penalty over 2+ minutes. - Yaw/altitude unaffected.

Caveat before relying on a vehicle: this is a real firmware change (it affects hardware, not just SITL). One **real-hardware** flight should confirm the SITL result before trusting it on an airframe — the sim's IMU noise/bias model is representative but not identical to the BMX160 + vibration environment.

Suggested commit message note: record the SITL validation (this archive + the 176 s drift check) and the residual-during-sustained-accel caveat, so the rationale travels with the change.

Open / out of scope (by decision)

5. **Sustained-acceleration tilt residual** ($\sim 8^\circ$ during a multi-second coordinated accel; p90 9° overall on aggressive flights) — the irreducible IMU-only floor. Removing it needs a **velocity reference** (GPS or optical-flow). A drag-model velocity estimate was explicitly **declined** (adds a calibration/modeling burden). So this residual is accepted: fine for manual/acro flight, and the bar for hands-off leveling / position hold would require the velocity sensor regardless.

Capture notes for next time

- **Log the active estimator config** (e.g. `EKF_R_ACC_DIR`, filter type) at arm so each run is self-describing — we currently track it out-of-band.
- For the **real-HW** check, fly a profile with a sustained coordinated acceleration (a fast, held lateral translation) — that's the regime that separates this estimator from a velocity-aided one.
- Keep using the dual-log (`export + gt-*.bin`) overlay as the regression baseline; this archive is the post-accel-trust reference.